

Introduction to Strings in Python

COVERS CREATION, MANIPULATION, BUILT-IN
FUNCTIONS, SLICING, AND MORE.



What is a String?



A string is a sequence of characters.



Enclosed in single (' '), double (" "), or triple quotes (''' '' or '''' ''').



Used to store and manipulate text in Python.

Creating Strings

Single quotes:
`s1 = 'Hello'`

Double quotes:
`s2 = "World"`

Triple quotes for
multiline: `s3 =
"Line1\nLine2"`

Traversing a String

Access characters
using indexing:
`s[0], s[-1]`

Use loops to
iterate: `for char
in string`

Multiline Strings



Use triple quotes to span across lines.



Useful for long text or documentation strings.

Concatenating Strings

Use + operator:

`'Hello' + ' ' + 'World'`

Repeat using *:

`'Hi' * 3 → 'HiHiHi'`

Escape Sequences



`\n`: New Line



`\t`: Tab



`\\`: Backslash



`\'`: Single Quote



`\"`: Double Quote

String Operators

+ :
Concatenation

*** :**
Repetition

in / not in :
Membership test

Common String Functions - 1

Function	Description	Example
<code>len(string)</code>	Length of string	<code>len("hello") → 5</code>
<code>string.upper()</code>	Convert to uppercase	<code>"abc".upper() → "ABC"</code>
<code>string.lower()</code>	Convert to lowercase	<code>"ABC".lower() → "abc"</code>
<code>string.endswith(value)</code>	Checks if ends with	<code>"hello".endswith("o") → True</code>
<code>string.startswith(value)</code>	Checks if starts with	<code>"hello".startswith("h") → True</code>
<code>string.find(value)</code>	Finds first occurrence	<code>"apple".find("p") → 1</code>
<code>string.islower()</code>	True if all lowercase	<code>"abc".islower() → True</code>

Common String Functions - 2

Function	Description	Example
<code>string.replace(old, new)</code>	Replaces substring	<code>"apple".replace("p", "b") → "abble"</code>
<code>string.count(value)</code>	Count occurrences	<code>"banana".count("a") → 3</code>
<code>string.isalpha()</code>	True if only letters	<code>"abc".isalpha() → True</code>
<code>string.isdigit()</code>	True if only digits	<code>"123".isdigit() → True</code>
<code>string.isalnum()</code>	True if alphanumeric	<code>"abc123".isalnum() → True</code>
<code>string.join(iterable)</code>	Joins iterable	<code>".".join(["a", "b"]) → "a.b"</code>

String Slicing

Syntax: `s[start:end:step]`

`s[::-1]` → reverse string

`s[::2]` → every 2nd character

String Examples

```
a = "Hello, World!"
print(a)
print(a[1])
print(len(a))

# characters by using the slice syntax
print(a[2:5])
print(a[2:])
print(a[:5])
print(a[-5:-2])
print(a.upper()) # uppercase
print(a.strip()) # removes any space at beginning,
print(a.replace("H", "J"))
print(a.split(",")) # returns ['Hello', ' World!']

# Looping through Strings:
for x in "banana":
    print(x)

age = 25
txt = f"My name is Xavi, I am {age}"
print(txt)
```

NUMBERS

```
x = 1    # int
y = 2.8  # float
z = 1j   # complex

# range(starting, ending + 1step, step)

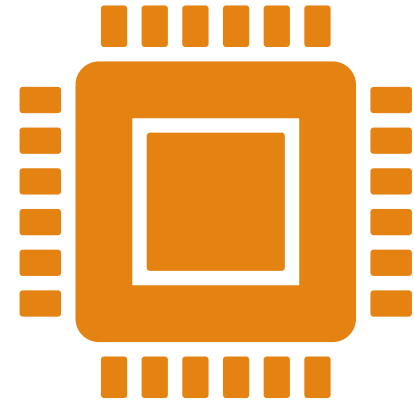
# Looping through Range:
for x in range(1, 10, 2):
    print(x)
```

Practice Questions

Count	Count number of vowels in a string.
Convert	Convert string to uppercase manually.
Check	Check if a string is a palindrome.
Replace	Replace spaces with hyphens.
Join	Join list ["Code", "With", "Python"] using space.

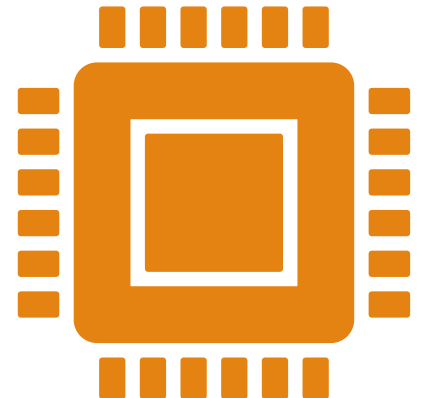
Quiz

1. What is the output of `"Hello"[1:4]`?
2. What does `"python".islower()` return?
3. How to check if string starts with 'Hi'?
4. What does `join()` do in strings?



Programs

1. Write a program that takes a string as input and checks if it's a palindrome (a word that reads the same backwards and forward). Ignore spaces and cases.
2. Write a program that takes a string as input and returns the count of vowels (a, e, i, o, u) in the string.



Programs

1. Write a program that takes a string as input and checks if it's a palindrome (a word that reads the same backwards and forward). Ignore spaces and cases.

```
s = input("Enter a string that you want to check: ")
s_lower = s.lower() # modifies string into lower case
s_no_space_lower = s_lower.replace(" ", "")
#removes all spaces from the string
j = s_no_space_lower[::-1]
# negative indexes to start from the end of the string
print("Is <" + s + "> palindrome?", s_no_space_lower==j)
```


Programs

2. Write a program that takes a string as input and returns the count of vowels (a, e, i, o, u) in the string.

```
string = input("Enter a string: ")
string = string.lower()
#Converted into lower case as 'I' and 'i' are considered different characters
vowels = "aeiou"
total = 0
for _ in vowels:
    count = string.count(_)
    total += count
print(total)
```